

A Critique and Enhancement Strategy for 'An Agent-Based Probabilistic Energy Management System'

Executive Summary of Critique and Strategic Recommendations

The manuscript 'An Agent-Based Probabilistic Energy Management System' presents a comprehensive and technically robust body of work. The research is timely, addressing critical challenges in the energy transition. The development of a dual-market framework, considering both the mature Dutch market and the transitioning Curaçao market, is a particularly novel and commendable aspect that demonstrates significant strategic foresight. The extensive simulations, detailed appendices, and the thoroughness of the technical implementation are clear strengths of the paper.¹ The core research is sound, and the results presented are compelling and significant.

The primary area for improvement lies not in the technical substance but in the narrative framing of the research. The current manuscript presents itself foremost as a report on an "agent-based" system. While the agent-based design is a sound software engineering choice that enables modularity and scalability, it is not the most profound scientific contribution of this work. The true novelty, which is currently understated, is the **adaptive optimization loop** at the heart of the system.

The central scientific innovation is the dynamic feedback mechanism where probabilistic models of user behavior, continuously learned from observed data, directly inform and adapt the Mixed-Integer Linear Programming (MILP) optimization. This is achieved by translating learned probabilities into a quantifiable penalty within the optimization's objective function. This creates a system that does not merely optimize based on static assumptions but actively learns and adapts its decision-making strategy to reflect human habits and preferences. This elevates the work from a description of a well-engineered system to the presentation of a new paradigm for human-centric energy optimization.

To maximize the impact of the manuscript, the following strategic recommendations are proposed:

1. **Refocus the Narrative:** The central story should be shifted from the "how" (the agent architecture) to the "what" (the adaptive probabilistic optimization framework). The title, abstract, and introduction should be revised to foreground this core contribution.
2. **Eliminate Redundancy and Improve Pacing:** The current structure with a separate Abstract, Executive Summary, and Management Introduction is repetitive and slows the reader's entry into the core content. These should be condensed into a single, powerful Introduction that immediately establishes the context, problem, and contributions.
3. **Elevate Key Concepts:** The analysis of the "cost of comfort"—the quantifiable trade-off between cost savings and user preference satisfaction—is a major finding that directly addresses a key challenge in the literature. This should be elevated from a side result to a central, user-centric feature and a cornerstone of the discussion.
4. **Integrate the Dual-Market Story:** The Netherlands/Curaçao context should be woven throughout the paper, not just mentioned in passing. It should be used to continuously reinforce the system's designed-in adaptability and strategic value, from the problem statement to the final conclusions.
5. **Restructure the Results Section:** The results should be organized around the key claims and insights the paper seeks to prove, rather than by a chronological reporting of simulation cases. This will create a more logical and compelling argument, transforming the section from a simple report of experiments into a validation of the paper's core thesis.

By implementing these changes, the manuscript can more effectively communicate the significance of its contributions, resulting in a more impactful and memorable paper. The following sections provide specific, insertable content and detailed suggestions to achieve this transformation.

Refining the Narrative and Introduction

To immediately improve the paper's focus and pacing, it is recommended to merge the existing Abstract, Executive Summary, and Management Introduction into a single, cohesive Introduction section. The Abstract should remain a standalone summary as

per standard publication formats, but the other two sections contain significant overlap that can be streamlined.

Proposed Unified Introduction

Modern energy systems are undergoing a rapid and fundamental transformation, driven by the increased penetration of intermittent renewable generation, the emergence of dynamic electricity pricing, and the proliferation of smart, controllable devices within buildings.¹ In advanced markets like the Netherlands, electricity prices can fluctuate on an hourly basis, creating significant opportunities for cost savings if consumption can be aligned with price signals. However, most buildings today lack the intelligent automation required to capitalize on this flexibility; appliances and HVAC systems are typically operated based on convenience or static schedules, with little regard for price variations. Consequently, building occupants often miss out on potential savings, and high peak demand continues to strain grid infrastructure.¹ Concurrently, the rise of intermittent renewables like solar PV and wind introduces new challenges of grid stability, as periods of surplus generation and deficit become more pronounced.¹

Energy Management Systems (EMS) for buildings have emerged as a promising solution to these challenges by intelligently coordinating the consumption and storage of electricity.¹ A building-level EMS can automatically schedule appliances, heating/cooling systems, and storage devices like batteries and Electric Vehicle (EV) chargers in response to price signals and renewable availability. For example, an EMS might delay EV charging from expensive evening hours to low-price overnight hours, provided user-defined constraints, such as having a full battery by morning, are met.¹ By orchestrating all flexible loads, an EMS can reduce the total energy bill for the end-user while simultaneously flattening the building's demand curve, which benefits the wider grid.

However, a critical review of existing literature reveals a key gap: most HEMS implementations treat user behavior as either deterministic or as a static probability distribution, failing to adapt to changing usage patterns over time. Furthermore, while many systems optimize for either cost or user comfort, few provide a systematic framework for balancing these often-competing objectives with tunable parameters. Finally, there is limited research on systems designed to function effectively across

different market structures, from advanced dynamic pricing to simpler tariff schemes.¹

This paper addresses these gaps by presenting the design, implementation, and validation of a novel, agent-based probabilistic Energy Management System. The central challenge addressed is the development of an EMS that not only delivers value in mature markets with existing dynamic pricing (e.g., the Netherlands) but is also strategically designed for phased deployment in emerging smart grid environments (e.g., Curaçao), where it must adapt from simple tariff structures to future dynamic ones.¹ The core innovation of this work is an

adaptive probabilistic scheduling framework. This framework integrates a machine learning component that continuously learns probabilistic models of user behavior with a Mixed-Integer Linear Programming (MILP) optimizer. This creates a dynamic feedback loop where the optimization strategy adapts over time to align with evolving human habits, systematically balancing cost reduction with user comfort.

The key contributions of this work are summarized in Table 1 and are substantiated in the subsequent sections of this paper.

Table 1. Key Contributions and Supporting Evidence

Contribution Claim	Brief Description	Substantiated in...
1. Adaptive Probabilistic Scheduling	A novel MILP formulation that integrates learned probabilistic models of user behavior as a dynamic penalty function, creating a feedback loop between user habits and optimization.	Methodology (§5.5, §5.7), Results (§7.3)
2. Quantifiable Comfort-Cost Framework	An explicit, tunable mechanism (wpref) that quantifies the economic trade-off between cost savings and user comfort, enabling personalized optimization.	Methodology (§5.5), Results (§7.3, Fig. V-30)
3. Modular, Agent-Based Architecture	A resilient and extensible five-layer software architecture that enables fault	Architecture (§4), Methodology (§5)

	tolerance and simplifies the integration of new devices and market signals.	
4. Dual-Market Design Philosophy	A flexible system design validated for both advanced (Netherlands) and transitioning (Curaçao) energy markets, ensuring broad applicability and future-proofing.	Introduction (§1), Results (§7.4), Business Case (§8.1)

The remainder of this paper is organized as follows: Section 2 reviews the state-of-the-art in building energy management. Section 3 details the technical architecture and mathematical formulation of the proposed EMS. Section 4 presents and discusses the evaluation results and performance metrics. Finally, Section 5 concludes with a summary of findings and directions for future work.

Fortifying the Technical Core: Architecture and Methodology

The technical sections of the paper are detailed and thorough. The following suggestions aim to sharpen the justification for key design choices and make the core technical innovation more explicit and unambiguous for the reader.

Justifying the Agent-Based Architecture

The "Project Ideation and Evolution" section provides an excellent chronological account of the decision to adopt an agent-based architecture.¹ To strengthen the paper, this justification should be moved from the "Ideation" section and integrated directly into the "System Architecture" section. This reframes the choice from a historical narrative ("we decided to...") to a deliberate design strategy ("The system is architected to achieve...").

Proposed Revised Opening for Section 4: System Architecture

The EMS is built upon a modular, multi-agent architecture designed to achieve three primary objectives: extensibility, resilience, and maintainability. This paradigm was chosen over a monolithic application structure to effectively manage the complexity of coordinating diverse energy assets and data sources. The agent-based approach provides several strategic advantages:

- **Modularity and Extensibility:** Each core function of the system (e.g., forecasting, optimization, device control) is encapsulated within an independent "agent" with well-defined interfaces. This modularity simplifies the integration of new technologies. For instance, adding a new type of controllable device, such as a smart thermostat or a pool pump, can be accomplished by developing a new corresponding agent with minimal impact on the existing system. This ensures the architecture is scalable and future-proof.
- **Resilience and Fault Tolerance:** In a multi-agent system, the failure of a single component does not necessarily lead to a total system failure. For example, if the agent responsible for retrieving weather forecasts is temporarily unavailable, the GlobalOptimizer agent can fall back to a default or historical forecast, allowing the system to continue operating in a degraded but functional mode. This fault isolation is critical for ensuring the reliability of a real-world energy management system.
- **Parallel Development and Maintainability:** The clear separation of concerns allows different development teams to work on separate agents concurrently. The ProbabilityModel agent, for instance, can be refined and improved independently of the GlobalOptimizer agent, as long as the data contract between them is maintained. This accelerates development cycles and simplifies long-term maintenance.

The system is organized into five primary layers, which house the various agents and define the flow of information... (The section would then continue with the description of the five layers as in the original manuscript).

Making the Probabilistic-Optimization Link Explicit

The most novel aspect of the methodology is the mathematical bridge between the machine learning models and the optimization engine. This link is described well in

prose across subsections 5.5 and 5.7, but its impact can be amplified by presenting it in a structured, unambiguous format. A dedicated table explicitly showing the data flow from the predictive agents into the MILP formulation would serve as a powerful technical centerpiece.

It is recommended to insert the following table in the Methodology section, immediately after the MILP objective function is presented. This will leave no doubt in the reader's mind as to how, precisely, the learned probabilities influence the final schedule.

Proposed Table: Model-to-Optimization Data Flow

Table 2. Information Flow from Predictive Agents to the MILP Formulation

Source Agent	Data Output	Data Type	Role in MILP	Corresponding Term in Formulation
GridAgent	Day-ahead prices	Vector (pt)	Parameter	Objective Function: $\sum pt \cdot Gt \Delta t$
PVAgent	PV generation forecast	Vector (PPV,t)	Parameter	Power Balance Constraint: $... = ... - PPV_t - ...$
ProbabilityModelAgent	User usage probability	Matrix (ProbPrefd,t)	Parameter	Objective Function: $\sum wpref(1 - ProbPrefd,t) x d,t$
BatteryAgent	Initial State of Charge	Scalar (SOCinit)	Parameter	SoC Update Constraint: $SOC_0 = SOC_{init}$
FlexibleDeviceAgent	Operational Constraints	Scalars (Ed,Ld,Δd)	Parameter	Device Scheduling Constraints

This table provides a clear and concise map of the system's intelligence. It demonstrates that the GridAgent and PVAgent provide external context (prices and weather-driven generation), while the FlexibleDeviceAgent and BatteryAgent provide

physical constraints.¹ Critically, it highlights the unique role of the

ProbabilityModelAgent: it provides an internal, learned model of human behavior that is directly integrated into the economic objective function of the optimizer.¹ This explicit link is the cornerstone of the system's adaptive capability.

Elevating the Results and Discussion

The Results section is comprehensive and presents a wealth of data from the simulations. To enhance its narrative impact, the structure should be reorganized to align with the key claims of the paper, rather than following the sequence of simulation cases. This transforms the section from a report of experiments into a compelling validation of the paper's core contributions.

Proposed Restructured Results and Discussion Section

7. Results and Discussion

This section presents the results of extensive simulations conducted to evaluate the performance of the proposed EMS. The experiments were designed to validate the system's core capabilities across a range of realistic scenarios. The discussion is structured to address the key research questions: the validation of core economic and grid benefits, the quantification of the trade-off between cost savings and user comfort, the system's adaptability to constrained environments, and its robustness under uncertainty.

7.1 Experimental Setup and Scenarios

(This subsection would remain largely the same, describing the simulation setup, data sources, and the different building scenarios as defined in the original manuscript.)

7.2 Validation of Core Economic and Grid Benefits

The primary objective of the EMS is to deliver tangible economic and grid-supportive benefits. Across all relevant scenarios, the system demonstrated significant performance improvements over the baseline (unoptimized) case.

- **Electricity Cost Reduction:** The EMS consistently reduced energy costs by intelligently shifting flexible loads to exploit price variations. For households with only schedulable loads and no on-site generation or storage, the system achieved average cost savings of 10-15%, with daily savings reaching as high as 36% on days with significant price volatility.¹ The introduction of battery storage amplified these savings considerably. By enabling energy arbitrage (charging during low-price periods and discharging during high-price periods), a battery increased total savings to the 30-50% range. In scenarios with both PV and battery storage, the system was often able to achieve near-zero or even negative net energy costs, effectively turning the building into a prosumer that could generate revenue from the grid.¹
- **PV Self-Consumption Increase:** For buildings with solar PV, the EMS proved highly effective at maximizing the on-site use of renewable energy. In a typical baseline scenario, PV self-consumption was around 40-50%, with the majority of solar energy being exported to the grid. The EMS, by scheduling controllable loads and battery charging to coincide with periods of high solar generation, increased the self-consumption rate to 80-95%.¹ This nearly doubles the utilization of on-site solar energy, directly reducing reliance on grid imports and improving the return on investment for the PV system.
- **Peak Demand Reduction:** The system's load-shifting and energy storage capabilities resulted in a significant reduction in peak demand from the grid. Even without a battery, intelligent scheduling of appliances reduced peak grid imports by an average of 20-30%.¹ When a battery was available, it could be used to supply power during peak hours, leading to peak demand reductions of over 50%. This peak-shaving capability not only reduces costs for consumers who may be subject to demand charges but also provides a valuable service to the grid by reducing system-wide stress.

7.3 The Cost of Comfort: Quantifying the Trade-off between Savings and User

Preference

A central challenge for any automated energy management system is gaining user acceptance. An overly aggressive optimization strategy that ignores user comfort is unlikely to succeed in a real-world deployment. The proposed EMS addresses this challenge directly through its probabilistic scheduling framework, which allows for an explicit and quantifiable trade-off between economic optimality and user convenience. This is controlled by the preference weight parameter, w_{pref} , which effectively acts as a user-configurable "comfort vs. savings" slider.

The analysis of this trade-off, presented in Figure V-30 and Table V-24 of the manuscript, reveals a critical non-linear relationship. When the system operates in a pure cost-optimization mode ($w_{pref}=0$), it achieves the maximum theoretical cost savings (averaging 32.7%) but at the expense of user comfort, with only 45% of appliance operations aligning with historical user patterns.¹ This could result in schedules that are economically optimal but practically inconvenient, such as running laundry at 3 AM.

However, as the preference weight is increased, user satisfaction rises sharply with only a modest reduction in savings. By setting w_{pref} to a moderate value (e.g., 3), the system achieves a high preference satisfaction rate of 87% while still delivering robust average cost savings of 25.4%.¹ This demonstrates that the EMS can capture the "low-hanging fruit" of user comfort—avoiding the most inconvenient scheduling decisions—at a minimal economic penalty. The cost of achieving this high level of comfort is approximately 7.3 percentage points of savings, a trade-off that is likely to be highly acceptable to most users.

This finding is significant because it provides a practical solution to the human-in-the-loop problem identified in the literature. It shows that it is not necessary to choose between maximum savings and user comfort; rather, a well-designed system can find a highly effective balance between the two. This tunable framework is a key feature for ensuring the long-term viability and adoption of the EMS.

7.4 Adaptability in Constrained Environments: The Low-Income and Emerging Market Case

To test the system's value proposition beyond high-tech, high-flexibility homes, the Res-L scenario was designed to represent a low-income household with minimal smart technology—only a single controllable water heater.¹ This scenario is also a proxy for the initial deployment phase in an emerging market like Curaçao, where smart device penetration is low and tariffs are not yet fully dynamic.

Even under these highly constrained conditions, the EMS delivered meaningful benefits. It achieved a 5-8% reduction in energy costs, which, while smaller in percentage terms than in other scenarios, translates to tangible absolute savings for a household facing energy poverty.¹ This level of savings is sufficient to provide a return on investment for a low-cost smart plug within a few months, suggesting that even minimal investment in smart technology can be economically rational when coupled with an intelligent control system.

When simulating a future transition to Time-of-Use (TOU) pricing, the savings in this scenario increased to 7-10%.¹ This result is particularly important as it demonstrates that the EMS can help vulnerable households benefit from, rather than be penalized by, the introduction of more complex tariff structures. This directly addresses energy equity concerns and highlights the system's role as an enabler of a just energy transition. The successful performance in this scenario validates the dual-market design philosophy, proving the system's ability to deliver value across a wide spectrum of economic and technological contexts.¹

7.5 Robustness and Performance under Uncertainty

A real-world EMS must be robust to the inherent uncertainties of energy forecasting. To evaluate this, additional experiments were conducted introducing random forecast errors of $\pm 20\%$ for both PV generation and load predictions. When the system operated with only a single day-ahead schedule, performance degraded moderately, with cost savings reduced by approximately 6.2% compared to the perfect-forecast case.¹

However, the system's architecture allows for intraday re-optimization. By implementing a single re-optimization at midday with updated information, the system recovered more than half of the performance loss, with the degradation in savings

reduced to just 2.8%.¹ This demonstrates that a simple and computationally inexpensive re-planning strategy is highly effective at mitigating the impact of forecast errors. The agent-based, modular design is what makes this adaptive re-planning feasible.

In terms of computational performance, the MILP optimization for a single building was consistently solved in under 10 seconds on standard hardware.¹ This is well within the practical limits for a day-ahead scheduling application and indicates that the approach is scalable for deployment across a large number of buildings.

Crafting a Powerful Conclusion and Future Work

The final sections of the paper should synthesize the findings and articulate their broader implications, leaving the reader with a clear understanding of the work's significance and its position within the field.

Proposed Revised Conclusion

This paper has presented an agent-based, probabilistic Energy Management System that successfully optimizes building-level energy use by integrating machine learning with mathematical optimization. The work's primary contribution is the development of an **adaptive probabilistic scheduling framework**, which creates a dynamic feedback loop between learned models of user behavior and the core optimization engine. This approach resolves a central tension in energy management: the conflict between mathematical optimality and human usability.

The results from extensive simulations demonstrate that this framework delivers significant, multifaceted benefits. It consistently reduces energy costs for consumers by 10-30% or more, nearly doubles the on-site utilization of solar PV generation, and reduces peak grid demand by 20-50%.¹ Crucially, these gains are achieved while maintaining a high degree of user comfort, with over 85% of operations aligning with user preferences. The system's "cost of comfort" analysis shows that this balance is achieved at a minimal economic penalty, providing a practical pathway for real-world

user acceptance.

Furthermore, the system's dual-market design philosophy, validated in both the advanced Dutch context and a simulated emerging market context for Curaçao, proves its adaptability and strategic value for a global energy transition. By delivering value even in highly constrained, low-income scenarios, the EMS demonstrates its potential to contribute to a more equitable energy future.

In conclusion, this work provides a robust blueprint for the next generation of smart energy management systems. By moving beyond static optimization and embracing an adaptive, human-centric approach, the proposed EMS turns buildings from passive energy consumers into active, intelligent participants in the energy ecosystem. It provides a viable path toward a more efficient, resilient, and sustainable energy system, delivering win-win outcomes for consumers, utilities, and society at large.

Proposed Refined Future Work

The future work section is well-considered. To further strengthen it, each proposed direction should be explicitly linked to a limitation or extension of the current work, demonstrating a clear and logical research trajectory.

Example Refinements:

- **Real-World Pilot Deployment:** While the simulations presented here provide a strong proof-of-concept, a limitation of this study is the absence of validation with physical hardware and real human interaction. The immediate next step is to conduct a pilot deployment in a small number of households. This will be critical for testing the robustness of the device integration layer, validating the user behavior models against real-time actions and overrides, and gathering qualitative feedback on the user interface and overall user experience.
- **Advanced Uncertainty Handling:** The current system mitigates forecast uncertainty through periodic re-optimization. While effective, this approach does not provide formal guarantees on performance. Future work will address this limitation by exploring more advanced uncertainty handling techniques. This includes the formulation of a **robust optimization** model, which would guarantee feasibility under worst-case forecast errors, and a **stochastic optimization** model, which would optimize the expected outcome over a range of probabilistic scenarios.¹ This would enhance the system's reliability, particularly for

applications requiring high-assurance performance.

- **Integration with Grid Services:** This research focused primarily on optimizing for the building owner's economic benefit. A natural extension is to enable the EMS to participate in broader grid-supportive activities. Future work will involve expanding the GridAgent to accept active demand response signals from a utility or aggregator. The GlobalOptimizer's objective function will be modified to co-optimize for both user cost and potential revenue from providing ancillary services, such as frequency regulation or emergency load reduction. This will require developing strategies for fair and effective aggregation across multiple EMS-equipped buildings.
- **Enhanced Economic and Lifecycle Analysis:** The current economic analysis focuses on operational savings. To provide a complete picture for potential adopters, future work will conduct a comprehensive lifecycle cost analysis. This will include the embodied carbon and capital cost of required hardware, the potential impact of increased cycling on battery degradation, and a detailed payback period analysis under various tariff structures and incentive programs. This will help to define clear business models for the commercial deployment of the EMS as a product or service.

Works cited

1. Main.tex

MORE

1. Logical Flow and Structure of the Paper

Introduction

The **Introduction** should clearly outline the research problem, the goals, and the motivations behind the study. Given that this is about energy management systems (EMS), focus should be on the following:

- **Research Problem:** Start by explaining the inefficiencies in current EMS, such as cost inefficiency, poor integration of renewable energy, and the lack of real-time adaptability.
- **Goals and Motivations:** Explain that the aim of the research is to create an agent-based EMS that can solve these inefficiencies through better decision-making, real-time data handling, and optimization models.
- **Transition to Methodology:** After the problem and goals are stated, naturally transition

into the methodology by explaining that the paper presents an agent-based architecture where different agents (battery, EV, PV) interact to optimize the energy usage of buildings.

Suggested Action:

- Start with a broader energy management problem and move towards the specific solution your research provides. Include terms like **cost reduction**, **real-time optimization**, and **renewable energy integration** in the introduction.

Methodology

The **Methodology** should follow the introduction, explaining the techniques and methods used. In your case, this is an agent-based approach and an optimization engine (likely MILP).

- **Agent-Based Approach:** Each agent (BatteryAgent, EVAgent, PVAgent, etc.) should be clearly defined with its specific role (e.g., EVAgent manages only charging, BatteryAgent has a full range of charge/discharge capabilities).
- **Optimization Engine:** Describe the MILP-based optimization process that ties these agents together and helps reduce energy costs.

Suggested Action:

- Introduce the agents first and explain their roles.
- Then, explain how the optimization engine works with these agents to minimize energy costs, improve renewable integration, and reduce peak loads.

Results

The **Results** should be clearly connected to the methodology:

- **Connection to Hypothesis:** Results like **cost savings**, **renewable energy utilization**, and **peak demand reduction** should directly reflect the success of the methodology outlined earlier.
- **Clear Evaluation:** Tie these results back to the initial problem. For example, if the cost savings are significant, explain how the optimization of energy use by the agents directly resulted in these savings.

Suggested Action:

- Reiterate the results from each agent's actions (e.g., the PVAgent's ability to reduce

energy costs, the EVAgent's efficient charging schedules).

Discussion

In the **Discussion**, interpret the results:

- **Comparison to Initial Goals:** How do the results validate the problem and solution presented in the introduction? For example, demonstrate how agent-based optimization directly improved cost savings and integration of renewable energy.
- **Implications:** Discuss the real-world impact of the proposed EMS. How can it be used in actual buildings or energy management systems?

Suggested Action:

- Draw connections between the results and real-world scenarios. Focus on scalability and adaptability of the proposed system.

2. Detailed Step-by-Step Review

Identifying Logical Gaps

- **Agent Responsibilities:** The paper should be very clear about which agents perform decision-making and which are just performing tasks. For example, the **GlobalOptimizer** makes high-level decisions, while agents like **PVAgent** only collect data.
- **Flow Between Sections:** There should be a smooth transition from problem introduction to methodology, and then to results. If a section introduces a concept without clear transitions (e.g., jumping from agent definitions to results), it could confuse readers.

Actionable Improvements:

- Revise the sections to ensure logical flow. Add transitions and explain how each agent's task directly contributes to the overall optimization.

Writing Style

The writing should be **clear and concise**. Avoid overly technical jargon without definitions and ensure that complex methodologies like MILP are explained in simple terms.

Actionable Improvements:

- Provide more explanations for terms like **MILP** and **agent-based modeling** for readers unfamiliar with these concepts.

3. Code Review

Code Alignment with Methodology

The code should strictly follow the methodology in the paper. Let's review whether the provided code matches the paper's described methodology.

- **GlobalOptimizer:** The **GlobalOptimizer** code should directly reflect the optimization techniques described. It should handle scheduling and control based on price signals, energy consumption, and other inputs. Ensure that the agents like **PVAgent**, **EVAgent**, and **BatteryAgent** are correctly instantiated and passed to the optimizer.
- **Code Optimization:** The `optimize()` function within **GlobalOptimizer** should be evaluated for performance. There are several places where redundant calculations could be simplified. Also, ensure the code aligns with the probabilistic modeling described in the paper.

Actionable Improvements:

- Optimize the flow in the `optimize()` function, particularly in the initialization of arrays and calculations.
- Ensure that the code comments explain the **why** behind each optimization step and link back to the methodology.

Redundancy and Efficiency

- Review the use of `deepcopy()` in the `monte_carlo_simulation` and `optimize()` functions. While it is necessary in some cases, it can lead to inefficiencies.

Actionable Improvements:

- Use more efficient ways of handling state duplication if possible, or explain why deep copies are required here.

Cross-Checking Results

- Cross-check the **savings** and other key results in the code against the results mentioned in the paper. Ensure that **savings** and other metrics like **battery state of charge** and **shifted energy loads** are correctly calculated.

Actionable Improvements:

- Check that the code accurately reports savings from each agent's actions and matches the results in the paper.

4. Actionable Improvements Summary

- **Introduction:** Strengthen the problem statement and transition into methodology. Clearly state the motivations and goals of the research.
- **Methodology:** Clarify agent roles and explain the optimization process in detail. Each agent's responsibilities must be clearly separated.
- **Results:** Tie each result back to the methodology and explicitly state how the agents' tasks lead to these outcomes.
- **Discussion:** Connect results to real-world applications. Explain the broader implications of your findings.
- **Code:** Improve code clarity, optimize performance, and ensure alignment with the theoretical methodology described in the paper.